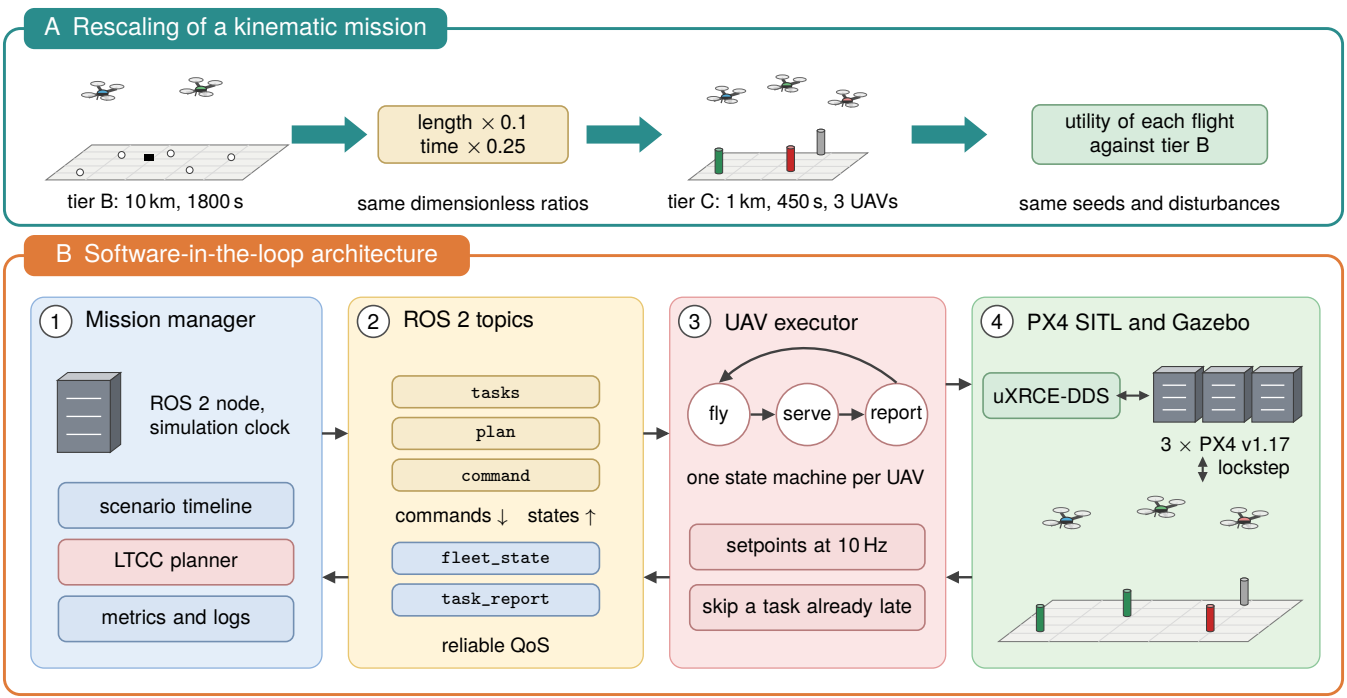

A Rescaling of a kinematic mission
tier B: 10 km, 1800 s
length × 0.1
time × 0.25
same dimensionless ratios
tier C: 1 km, 450 s, 3 UAVs
utility of each flight against tier B
same seeds and disturbances
B Software-in-the-loop architecture
1 Mission manager
ROS 2 node, simulation clock
scenario timeline
LTCC planner
metrics and logs
2 ROS 2 topics
tasks
plan
command
commands ↓ states ↑
fleet_state
task_report
reliable QoS
3 UAV executor
fly
serve
report
one state machine per UAV
setpoints at 10 Hz
skip a task already late
4 PX4 SITL and Gazebo
uXRCE-DDS
3 × PX4 v1.17
lockstep